

Logic Gates

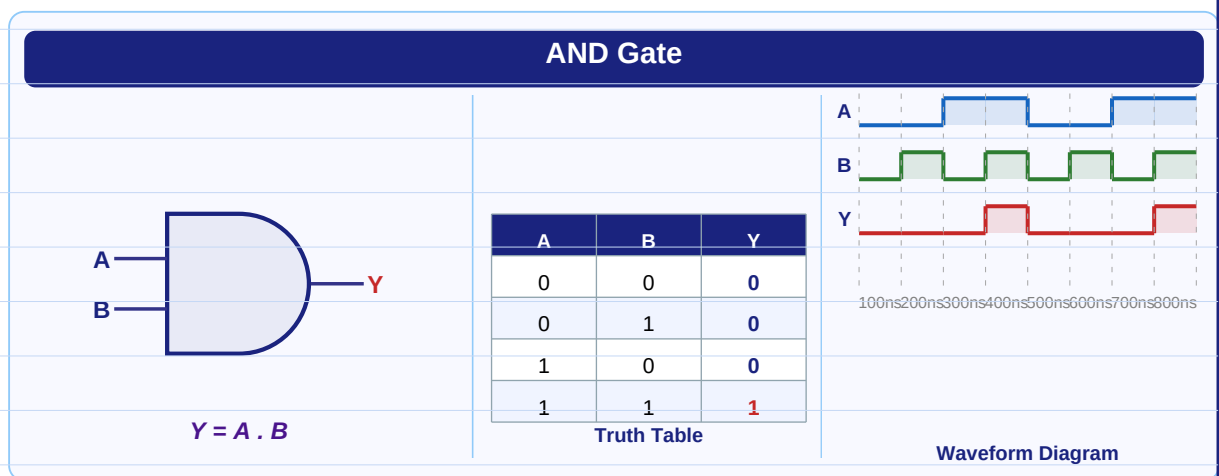
Gate Symbols, Truth Tables & Waveform Diagrams

This document presents gate symbols, truth tables, and waveform diagrams for all fundamental and universal logic gates used in digital electronics.

1. AND Gate

Theory: AND gate performs logical multiplication. Output is HIGH (1) only when ALL inputs are HIGH (1). Used in control circuits.

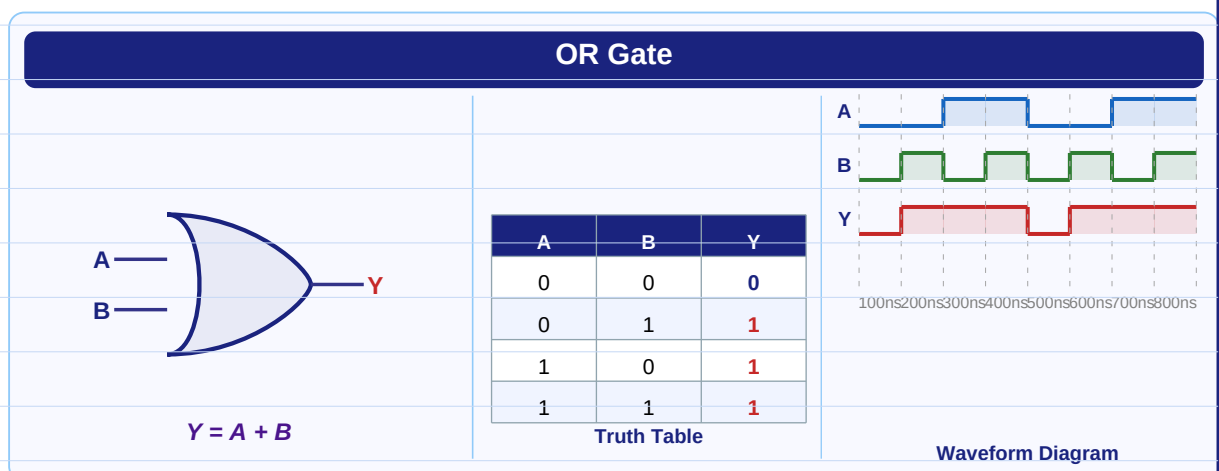
Expression: $Y = A \cdot B$



2. OR Gate

Theory: OR gate performs logical addition. Output is HIGH (1) when ANY input is HIGH (1). Output is LOW only when all inputs are LOW.

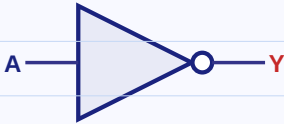
Expression: $Y = A + B$



3. NOT Gate

Theory: NOT gate (inverter) reverses the input signal. If input is HIGH, output is LOW and vice versa.
Single input, single output.

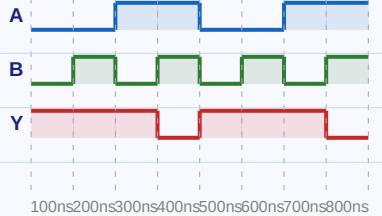
Expression: $Y = A'$

NOT Gate								
 $Y = A'$	<table border="1"> <thead> <tr> <th>A</th> <th>Y</th> </tr> </thead> <tbody> <tr> <td>0</td> <td>1</td> </tr> <tr> <td>1</td> <td>0</td> </tr> </tbody> </table> Truth Table	A	Y	0	1	1	0	 Waveform Diagram
A	Y							
0	1							
1	0							

4. NAND Gate

Theory: NAND is a universal gate = AND + NOT. Output is LOW (0) only when ALL inputs are HIGH.
Can implement any logic circuit.

Expression: $Y = (A \cdot B)'$

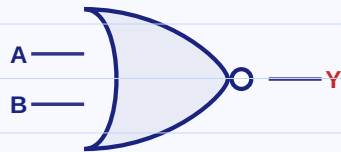
NAND Gate																	
 $Y = (A \cdot B)'$	<table border="1"> <thead> <tr> <th>A</th> <th>B</th> <th>Y</th> </tr> </thead> <tbody> <tr> <td>0</td> <td>0</td> <td>1</td> </tr> <tr> <td>0</td> <td>1</td> <td>1</td> </tr> <tr> <td>1</td> <td>0</td> <td>1</td> </tr> <tr> <td>1</td> <td>1</td> <td>0</td> </tr> </tbody> </table> Truth Table	A	B	Y	0	0	1	0	1	1	1	0	1	1	1	0	 Waveform Diagram
A	B	Y															
0	0	1															
0	1	1															
1	0	1															
1	1	0															

5. NOR Gate

Theory: NOR is a universal gate = OR + NOT. Output is HIGH (1) only when ALL inputs are LOW (0).
Can implement any logic circuit.

Expression: $Y = (A + B)'$

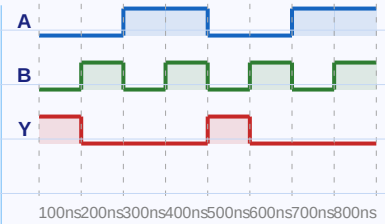
NOR Gate



$$Y = (A + B)'$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Truth Table



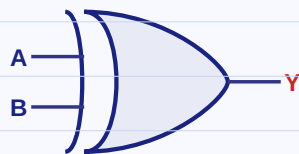
Waveform Diagram

6. XOR Gate

Theory: XOR (Exclusive-OR) gives HIGH (1) when inputs are DIFFERENT. Output is LOW when both inputs are same. Used in adder circuits.

Expression: $Y = A \text{ XOR } B$

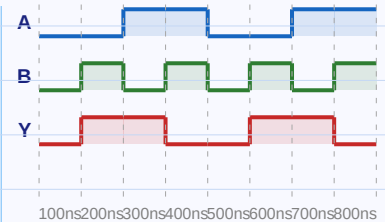
XOR Gate



$$Y = A \text{ XOR } B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Truth Table



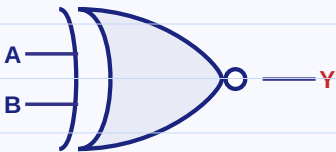
Waveform Diagram

7. XNOR Gate

Theory: XNOR (Exclusive-NOR) gives HIGH (1) when inputs are SAME. Also called equivalence gate. Output is LOW when inputs differ.

Expression: $Y = (A \text{ XOR } B)'$

XNOR Gate



$Y = (A \text{ XOR } B)'$

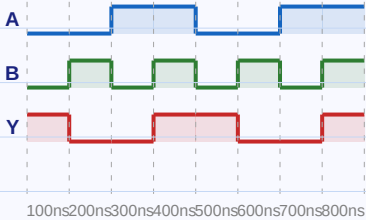
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Truth Table

A

B

Y



100ns200ns300ns400ns500ns600ns700ns800ns

Waveform Diagram